

EZHILAN CHINNASAMY

ezhilan03@gmail.com | +1 4693326833 | Dallas, Texas, US | Open to UK relocation
LinkedIn | GitHub | Portfolio

Data engineer building reliable payments pipelines and analytical tools. Designed data infrastructure processing 2M records daily and led integrations across three teams, connecting hands-on delivery with operational improvements.

SKILLS

- **Data Engineering:** ETL/ELT design, star-schema modelling, Airflow orchestration, data lakes, API/SFTP ingestion
- **Languages & Platforms:** SQL, Python, PostgreSQL, Azure Data Factory, AWS S3/EMR, PySpark
- **Analytics & Quality:** Financial reconciliation, evaluation harnesses, Grafana business-operations monitoring
- **Engineering Delivery:** dbt, BigQuery, Docker, Terraform, GitHub Actions, CloudWatch (portfolio)

PROFESSIONAL EXPERIENCE

Pay with Spire, Business Intelligence Analyst Sep 2024 - Jun 2026 | Dallas, Texas, US

- Designed a bronze/silver/gold Azure data lake with Azure Data Factory pipelines processing 2M records daily for multi-team reporting, marketing automation and AI analytics.
- Led three-team delivery of settlement automation, translating payment-processing rules into domain architecture and coordinating vendor API/SFTP integration across operations and engineering.
- Built modular Python merchant onboarding with partner-file validation, API integration and regression tests, saving an estimated 8.5 hours of manual entry per week.
- Built a production multi-agent analytics layer over the warehouse using Google ADK and Vertex AI, enabling self-service NL2SQL/BQML querying and offloading approximately 30% of data-team workload.

The University of Texas at Dallas, Student Assistant Jun 2024 - Sep 2024

- Automated public-record integration and trained 50+ student researchers, accelerating project delivery by 30%.

Ernst & Young Global Delivery Services, Software Engineer Feb 2021 - Oct 2022 | Chennai, India

- Increased batch throughput by 24% with PySpark pipelines on AWS EMR processing multi-terabyte logs.
- Architected a secure migration framework from legacy multi-tenant databases to an AWS S3 data lake, strengthening IAM controls and improving query performance by 35%.

PROJECTS

Underwriting / Payment Risk - v0.1.0

- Engineered point-in-time features for 3,600 synthetic applications; verified five dbt models and 17 tests on BigQuery Sandbox, including source restoration. Model training runs locally.

Fitness Data Platform - v1.0.0

- Developed versioned synthetic workout ingestion and dbt summaries; verified Airflow retries and backfills, on-demand AWS Fargate delivery and IAM-authenticated API access.

Multi-Agent Financial Reconciliation Engine

- Implemented replay-safe matching and human-review routing; AWS demo matched 383 of 500 synthetic transactions, with 117 retained for review and backup restoration verified.

Fintech Hybrid RAG Engine - v0.2.0

- Delivered BM25/pgvector retrieval with citations and abstention; verified 84 tests and on-demand AWS delivery, backup restoration and dependency recovery.

Revenue Pipeline Reliability - local release

- Orchestrated PostgreSQL/dbt validated publications, Airflow orchestration and checksum-verified replay; verified 106 PostgreSQL-backed tests and backup restoration. Azure deployment remains deferred.

EDUCATION & CERTIFICATIONS

The University of Texas at Dallas, MS in Business Analytics and Artificial Intelligence Jan 2023 - Dec 2024 | GPA: 3.7

Certifications issued by UTD: Applied Machine Learning Certificate

Kumaraguru College of Technology, Bachelors in Computer Science and Engineering Jul 2017 - Jun 2021 | GPA: 8.1

Scrum Master Accredited Certification - International Scrum Institute | Mar 2026